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2. A has a pivot in every row
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9. T is injective
10. T is surjective

We have the tools to prove all these, but really, it's just a summary of all the linear transformation and matrix algebra we've done so far!

One-sided inverse Implies the Other!

Theorem

Let $A \in \mathbb{R}^{n \times n}$. If there is some $B \in \mathbb{R}^{n \times n}$ such that

$$BA = I_n \text{ or } AB = I_n$$

Then, A is invertible and $A^{-1} = B$.

$AB = I_n$ case

Proof.

First, let $B \in \mathbb{R}^{n \times n}$ such that $AB = I_n$.

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We will show that T is surjective by showing how we can get any $\mathbf{b} \in \mathbb{R}^n$.

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Since we have an input, $\mathbf{x} = B\mathbf{b}$ such that $T(\mathbf{x}) = \mathbf{b}$,

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Assume that $\mathbf{x} \in \mathbb{R}^n$ such that $A\mathbf{x} = \mathbf{0}$.

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Assume that $\mathbf{x} \in \mathbb{R}^n$ such that $A\mathbf{x} = \mathbf{0}$. See that

$$\begin{aligned} A\mathbf{x} &= \mathbf{0} \\ BA\mathbf{x} &= B\mathbf{0} \\ I_n\mathbf{x} &= \mathbf{0} \\ \mathbf{x} &= \mathbf{0} \end{aligned}$$

So, if $A\mathbf{x} = \mathbf{0}$, then $\mathbf{x}$ must be $\mathbf{0}$!

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